LAB SHEET – 10

SMART CAMPUS LOST & FOUND MANAGEMENT SYSTEM

Course: BCA I Semester
Session: 2026–2027
Programming Language: C++

Submitted By: _____

Roll Number: _____

Submitted To: _____

Copy this report neatly into your practical notebook if a handwritten report is required.

1. Objective

To develop a system where students can report lost items, submit found items, search for belongings, and match records automatically.

2. Problem Statement

Every college faces the problem of lost items such as ID cards, wallets, keys, laptops, and mobile phones. This system makes finding lost items faster and more organized.

3. Software Requirements

1. C++ programming environment.
2. Text editor or IDE.
3. Computer system.

4. Modules / Features

No.	Module	Description
1	Student Login	Accepts student name and roll number.
2	Report Lost Item	Records details of an item a student has lost.
3	Report Found Item	Records details of an item a student has found.
4	Search Item	Searches by item name, category or location.
5	Auto Match Lost & Found	Finds possible matches by item name and category.
6	View All Records	Displays all records in alphabetical order by item name.
7	Delete Record	Allows a student to delete their own record by ID.
8	Item Statistics	Displays total, lost and found record counts.
9	Export Report	Creates a text report containing all records.
10	Exit	Closes the program.

5. Concepts Used

Arrays, structures, file handling, functions, loops, searching, sorting and strings.

6. Algorithm / Logic

• Start the program and load saved item records from lost_found.txt.

• Ask the student to enter their name and roll number for login.

• Display the menu after successful login.

• To report a lost or found item, accept item name, category, location, date and description.

• Assign a unique record ID, store the record in an array and save all records in the file.

• Compare a new item with existing records of the opposite type using item name and category. Display possible matches.

• Search records using item name, category or location.

• Display all records after sorting them alphabetically by item name.

• Allow a student to delete their own record using its ID.

• Display total, lost and found record counts.

• Export all records to lost_found_report.txt when requested.

• Repeat the menu until Exit is selected.

7. Auto Matching Logic

The system compares a LOST record with a FOUND record. If item names and categories are the same (ignoring uppercase and lowercase letters), it displays a possible match. This is only a suggestion; ownership must be verified by the student or campus office.

8. File Handling

lost_found.txt: Stores ID, student name, roll number, item name, category, location, date, description and type (LOST/FOUND).
lost_found_report.txt: Readable exported report containing all records.

9. Sample Input and Output

```
SMART CAMPUS LOST & FOUND MANAGEMENT SYSTEM
Student name: Rahul Sharma
Roll number: BCA101
Login successful. Welcome, Rahul Sharma!

Example: Report LOST item
Item name: Black Wallet
Category: Wallet
Location: Library
Date: 2026-11-01
Description: Black leather wallet
Report saved.

Then report FOUND item with:
Item name: Black Wallet
Category: Wallet
Location: Library Desk
Date: 2026-11-01
Description: Black wallet found near desk

POSSIBLE MATCH FOUND!
Please verify ownership with the campus office.
```

Note: The sample is illustrative. Enter the same item name and category in the LOST and FOUND records to demonstrate matching.

10. Conclusion

The Smart Campus Lost & Found Management System provides an organized way to report lost and found belongings. It supports searching, possible automatic matching, viewing and sorting records, deleting a student's own record, displaying statistics and exporting reports. It demonstrates practical C++ programming concepts and file handling.

11. Future Enhancements

QR Code Verification, Email Notification, Face Recognition and Mobile Application.

12. Important Demonstration Notes

Use the same item name and category in LOST and FOUND reports to demonstrate auto matching. Searching ignores uppercase/lowercase and can find part of an item name, category or location. Records are saved in lost_found.txt in the program's working folder. Export creates lost_found_report.txt. Student login by name and roll number is a classroom demonstration, not secure authentication.

Appendix: Source Code

The complete C++ source code is supplied separately as Smart_Campus_Lost_and_Found.cpp. Attach or print that file as the source-code section required by the lab sheet.